

Sandesh Kumar Gupta

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Summary

Pre-final year Computer Science student specializing in Cybersecurity, Systems Programming, and Linux Kernel Security. Experienced in developing eBPF based security systems, network detection, and vulnerability assessment and penetration testing.

Education

GLA University B.Tech, Computer Science and Engineering CGPA: 7.08/10	Mathura, Uttar Pradesh 2024 – 2028
Surmount International School Class XII — 76.40% Class X — 94%	Gorakhpur, Uttar Pradesh 2023 2021

Experience & Security Research

Cyber Security Intern — APCSIP-2026 2026 – Present
Amroha Police Cyber Security Internship Program

- Assisted in malware analysis, digital forensics, incident response exercises, and vulnerability assessment during the Amroha Police Cyber Security Internship Program.
- Performed vulnerability assessment & penetration testing (VAPT), producing structured research reports on identified CVEs and remediation strategies.

Independent Security Researcher & CTF Competitor 2024 – Present

- 7th / 9,000+ teams** — Hack IITK CTF (C3iHub, IIT Kanpur, 2026): Reverse Engineering, OSINT, Web Exploitation, ICS/SCADA, Cryptography.
- Top 2% globally** on TryHackMe; Top 20 across national & international CTF competitions.

Projects

eRDS — eBPF Ransomware Defense System 2026 · GitHub

- Developed an eBPF CO-RE kernel module to monitor `sys_enter_write` events with sub-millisecond latency.
- Implemented kernel-level ransomware detection using Linux tracepoints and ring buffers, portable across Linux 5.8+ without recompilation.
- Designed a three-tier entropy-based classification engine in **Go** with automated ransomware process termination for ransomware mitigation.

PacketDive — Network Detection & Response Engine 2026 · GitHub

- Built a Python NDR engine using Scapy and producer-consumer threading, achieving sustained packet capture without observed packet loss during benchmark testing.
- Reduced traffic noise by 30–40% via kernel-level BPF pre-filtering before user-space processing.
- Designed and implemented an **Isolation Forest** based anomaly detection pipeline for real-time network traffic analysis, exporting structured alerts compatible with SIEM platforms.

Technical Skills

- Languages:** Python, C++, Java, JavaScript, Bash
- Cybersecurity:** Burp Suite, Metasploit, Nmap, Wireshark
- Systems & Platforms:** Linux Kernel, Linux (Ubuntu, Kali), eBPF, Docker, Git, Github
- Web:** HTML, CSS, React
- AI & Concepts:** LLM Security, Prompt Injection, Threat Detection, Incident Response

Certifications

Google Cybersecurity Professional Certificate — Google | **Intro to Penetration Testing** — Security Blue Team | **Certified LLM Security Professional (CLLMSP)** — Red Team Leaders | **AI for Cybersecurity** — LinkedIn Learning | **Certified Online Fraud Prevention Specialist (COFPS)** — Hack & Fix Academy | **Full Stack Web Dev** — Internshala

Achievements

APCSIP-2026 — Amroha Police Cyber Security Intern | **7th / 9,000+ Teams** — Hack IITK CTF, IIT Kanpur (2026) | **Top 2% Globally** — TryHackMe (3M+ users) | **5th Place** — Scythe CTF, IIT Roorkee (2026) | **Top 30** — CodeFest CTF (IIT BHU) & Bypass CTF